Machine Learning Coursework 1

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1 Exploratory Data Analysis

The dataset consists of 10 000 rows and 31 columns: one continuous outcome variable and 30 diamond-related features (3 categorical and 27 numerical). There were no missing values or duplicates.

1.1 Distribution of Numeric Features

Numeric features consist of x, y, z, price, carat, depth, table, a1–a10 and b1-b10.

The outcome variable has a normal distribution and did not require transformation. However, the y column showed extreme skewness (8.37) with a maximum of 58.9, a median of 5.72 and minimum of zero. Since diamonds cannot have zero dimensions, and almost all values are clustered between 0 and 10, the maximum must be an outlier. Both x and z also contain minimum values of 0, which are impossible and represent data errors.

Price (skew = 1.63) and carat (skew = 1.15) are significantly right-skewed, which is expected since most diamonds are small and cheap. The features x, z, and depth are moderately skewed. The a/b features have approximately zero skew and can be categorised into two groups: those with uniform distribution and those with a normal distribution.

Log transformations were tested on the skewed features, but this did not improve linear model performance and was therefore not included in the final preprocessing pipeline. Tree-based models are invariant to monotonic transformations of features, so this did not affect their performance.

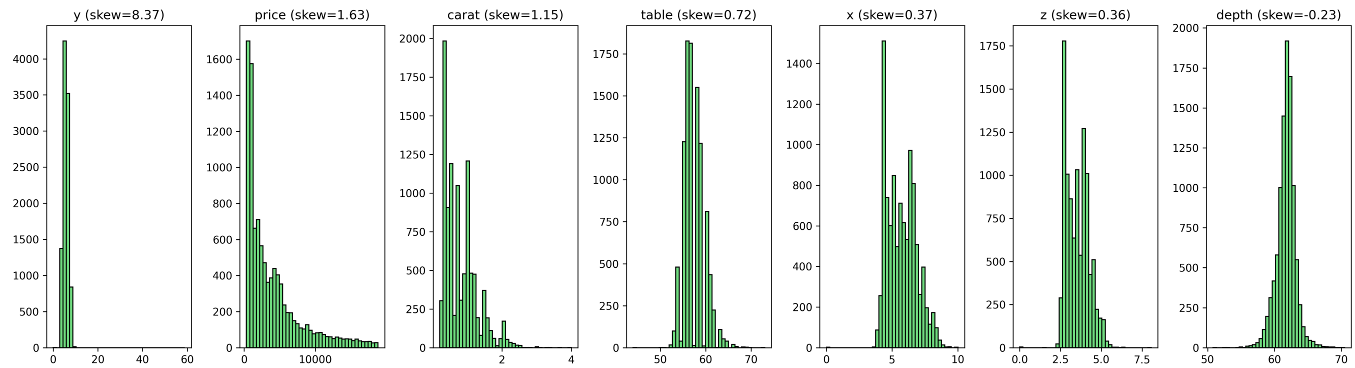

Figure 1: Feature Distribution

1.2 Categorical Features

The categorical features (cut, color, clarity) each showed class imbalance, which reflects the real-world data where diamonds with better cut and clarity are more common. This imbalance means that models may perform better on the more common categories.

1.3 Correlation with Outcome

Most features have a weak linear relationship with the outcome, suggesting that tree-based models will perform better than linear models. With Pearson correlation, it was found that depth has the strongest linear relationship (r = -0.41), followed by b3 (r = 0.23) and b1 (r=0.17). Despite the a/b features having weak individual correlations, removing them was detrimental to model performance, which indicates that they capture non-linear relationships with the outcome.

1.4 Multicollinearity

The features carat, price, x, y, and z are all correlated with $r \geq 0.8$, which is expected since larger diamonds have larger dimensions and cost more. For example, x and z are almost perfectly correlated (r = 0.99). A few a/b features showed moderate within-group correlation.

Dropping the highly correlated redundant features (price, x, y, z) had negligible impact on GBM performance (R^2 decreased from 0.4757 to 0.4755), which is why it was not included in the data preprocessing.

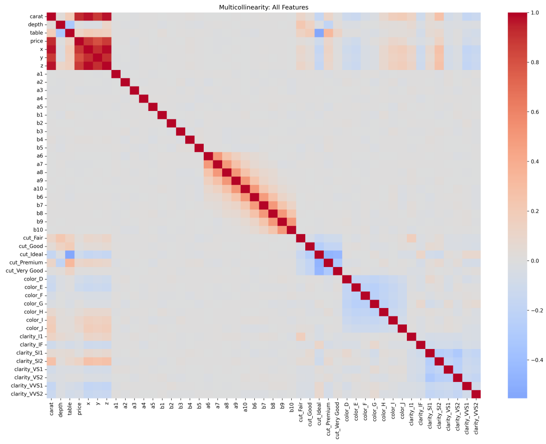

Figure 2: Feature Correlation

1.5 Data Preprocessing

A shared preprocessing pipeline was applied across all models:

- Removal of 5 invalid rows: 4 rows with zero dimensions for x, y, or z, and 1 row with an extreme y outlier.
- One-hot encoding of categorical features with drop_first=True.
- Standardisation of all features, which is essential for regularised linear models and does not affect tree-based models.

2 Model Selection

Five different model types were compared, which consists of linear models, a single decision tree and three ensemble methods.

2.1 Regularised Linear Regression

Ridge, Lasso and Elastic Net regression were included as baseline models. Given the weak linear correlations observed between the features and the outcome, these were not expected to perform well. Lasso performs automatic feature selection by shrinking coefficients via an L1 penalty, ridge handles multicollinearity with an L2 penalty, and Elastic Net combines both.

2.2 CART (Decision Tree)

A single Classification and Regression Tree was included as the baseline for tree-based methods. CART captures non-linear relationships and feature interactions, but it is prone to high variance and overfitting, typically resulting in poor generalisation.

2.3 Random Forest

Random Forest improves upon CART by combining multiple trees via bagging. Each tree is trained on a different random sample of the training data, and uses a random subset of features at each split. This decorrelates the trees and reduces variance.

2.4 AdaBoost

AdaBoost was included to compare the boosting ensemble approach against bagging. It fits shallow decision trees sequentially, reweighting training samples so that subsequent trees focus on previously misclassified examples. AdaBoost tends to be outperformed by gradient boosting on regression tasks and is more sensitive to noise.

2.5 Gradient Boosting Machines (GBM)

GBM was expected to be the best-performing model. Like AdaBoost, it fits trees sequentially, but each new tree is trained on the residuals of the previous ensemble. GBMs are recognised to perform well on tabular regression tasks. The main risk is overfitting, but this is mitigated through cross-validation.

3 Model Training

3.1 Training Procedure

All models were evaluated using 5-fold cross-validation with R^2 as the scoring metric. Most models used GridSearchCV, which searches all hyperparameter combinations. GBM used RandomizedSearchCV instead, which samples a fixed number of random combinations, as the alternative was computationally expensive. In both cases, the best-performing combination is selected and the model is trained on the full training set.

3.2 Model Evaluation

The mean R^2 values across 5 folds:

Model	$\mathbf{R^2}$	**Best Hyperparameters**
Ridge	0.2857	alpha = 100
Lasso	0.2865	alpha = 0.1
Elastic Net	0.2864	alpha = 0.1, l1_ratio = 0.9
CART	0.4221	max_depth = 5, min_samples_leaf = 5, min_samples_split = 2
Random Forest	0.4594	max_depth = 10, min_samples_leaf = 10, n_estimators = 500
AdaBoost	0.4615	estimator__max_depth = 7, learning_rate = 0.05, n_estimators = 500
GBM	0.4757	subsample = 0.7, n_estimators = 800, min_samples_leaf = 20, max_depth = 3, learning_rate = 0.01

Table 1: Model Performance and Best Hyperparameters

The linear models performed the worst, confirming that the data contains mostly non-linear relationships. Regularisation provided only a small improvement over plain linear regression, with Lasso slightly outperforming Ridge and Elastic Net.

Among the tree-based models, CART was outperformed by all ensemble methods, with GBM achieving the best score of 0.4757. GBM was therefore selected as the final model for generating test set predictions.

3.3 Feature Importance

The GBM importance feature ranking confirmed that depth was the most important predictor, which is consistent with it having the strongest linear correlation with the outcome. This was followed by b3, b1 and a1.

4 Code Supplement

The full implementation is available at: `https://github.com/katiepreed/COURSEWORKML`. Each model has its own script that performs hyperparameter tuning via cross-validation. Visualisations of the data analysis are saved to the plots directory and the final GBM predictions are saved to CW1_submission_K21008618.csv.